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Data 2024/11/05	Territoriality Wu Hu	Version 1.0	Page 3 / 12

Contents

No.		Page
	REVISION HISTORY	2
	CONTENTS	3
1.0	SCOPE	4
2.0	PURPOSE	4
3.0	PERSONNEL	4
4.0	GENERAL SPECIFICATION	4~5
5.0	Panel Outline	6
5.1	TFT Sensor Panel Outline	6
5.2	Panel After Bonding Outline	6
6.0	14 Layers TFT Process of 3.99 inch TFT sensor	7
7.0	Electrical Diagram of 3.99 inch TFT sensor	7
7.1	Electrical Input Range of 3.99 inch TFT sensor	8
8.0	TEG SPECIFICATION of 3.99 inch TFT sensor	9
9.0	TEST SPECIFICATION	10
9.1	Defect Definitions	10
9.2	Defects Specifications	10
10.0	VISUAL INSPECTION & AOI INSPECTION	11
11.0	IC PAD & COF PIN ASSIGNMENT	12
11.1	Pad Positions and Pin Assignment	12
11.2	ROIC COF Dimension and Pin Assignment	TBD
11.3	Gate COF Dimension and Pin Assignment	TBD
12.0	ENVIRONMENTAL SPECIFICATION	TBD
12.1	Electrostatic Discharge Sensitivity	TBD
13.0	PRODUCT IDENTIFICATION AND PACKAGE	TBD
13.1	Array Identification	TBD
13.2	Array Test Documentation	TBD
13.3	Array Packaging	TBD
13.4	Packing Test Conditions	TBD
13.5	Box & Tray packaging	TBD
13.6	Barcode & mark for history	TBD

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Data 2024/11/05	Territoriality Wu Hu	Version 1.0	Page 4 / 12

1.0 SCOPE

This document provides the product requirement for the 3.99 inch TFT sensor, 7.168 cm(x) * 7.168 cm(y) metal oxide TFT (IGZO TFT) sensor array. The process flow of 3.99 inch TFT sensor is 14 photolithography and etching process (PEP). The array panel is composed of a glass substrate on which materials are deposited to fabricate the thin film transistor (TFT), PD sensor layer, Bias lines and contact pads, those enable the connection of the pixel array to the readout electronics.

2.0 PURPOSE

This document provides the product requirements for the 3.99 inch TFT sensor, 7.168 cm(x) * 7.168 cm(y) sensor array.

3.0 PERSONNEL

This document applies to customers with external supplier on the 3.99 inch TFT sensor.

4.0 GENERAL SPECIFICATION

The DTF 3.99 inch TFT sensor array is a two-dimensional array with thin-film transistor switches based on metal oxide (IGZO) technology. The array has a pixel area of 7.168 cm(x) * 7.168 cm(y) . Pixels are addressed through 512 data lines and 512 gate lines.
The 262,144 pixels are arranged in a matrix of 512 (d) x 512(g). Each pixel size is 140 μm * 140 μm. The table below defines the result of assessment and the design parameter of 3.99 inch TFT sensor array

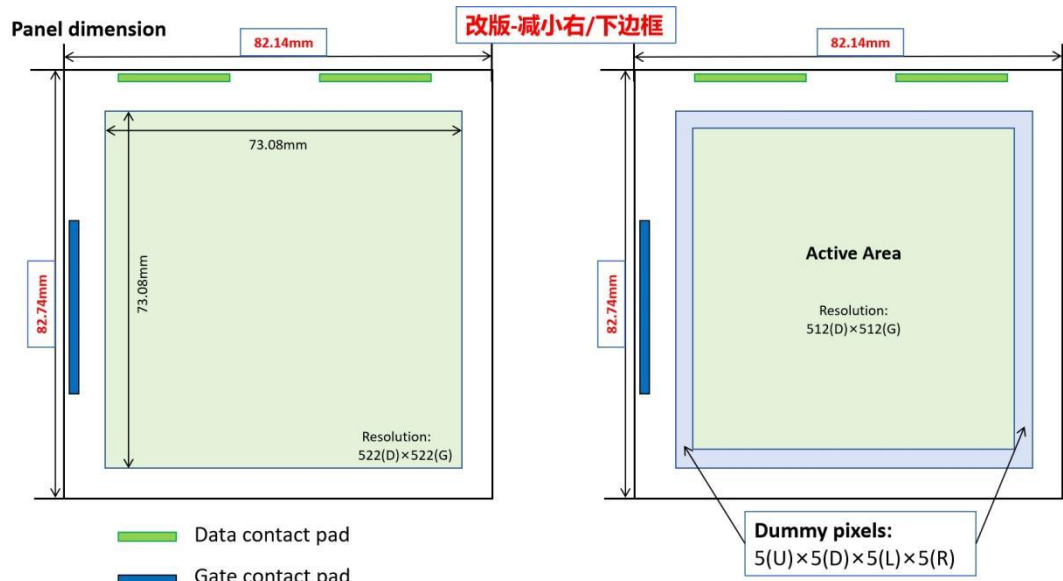
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Data 2024/11/05	Territoriality Wu Hu	Version 1.0	Page 5 / 12

	Design parameter	Specifiaction
Basic Spec.	Open cell size (outer COF) (H*V)	118.740mm × 118.140mm
	Panel size (H*V)	82.14mm × 82.74mm
	Active pixel area size (H*V)	71.68mm × 71.68mm
	Number of active area pixel (H*V)	512×512=262144
	Number of data line	512
	Number of gate line	512
	Panel thickness	0.70mm
	Open cell thickness(with COF + CsI)	1.58mm
	Open cell weight(with COF + CsI)	≈25g
Pixel spec.	Pixel pitch	140um×140um
	TFT channel size	W/L 7.44/8.4
	Gate line width	5.5um
	Gate line material	Al \ Mo
	Data line width	7.5um
	Data line material	Mo \ Al \ Mo
	Bias line width	4.6um
	Bias line material	Mo \ Al \ Mo
	PD sensor material	N+ \ a-Si \ P+
Peripheral spec.	Number of Gate Pad Blocks	1 block
	Number of Data Pad Blocks	2 block
	Gate Pad Block Pitch	/
	Data Pad Block Pitch	35.84mm
	Gate Pads per Block	532 per block
	Data Pads per Block	270 per block
	Contact Pad Metal	Mo \ Al \ Mo\ITO
	Contact Pad Pitch(Gate, Data)	72 um / 100um
	Contact Pad Width(Gate, Data)	44 um / 66 um
	Contact Pad Length(Gate, Data)	1.0 mm / 2.2mm

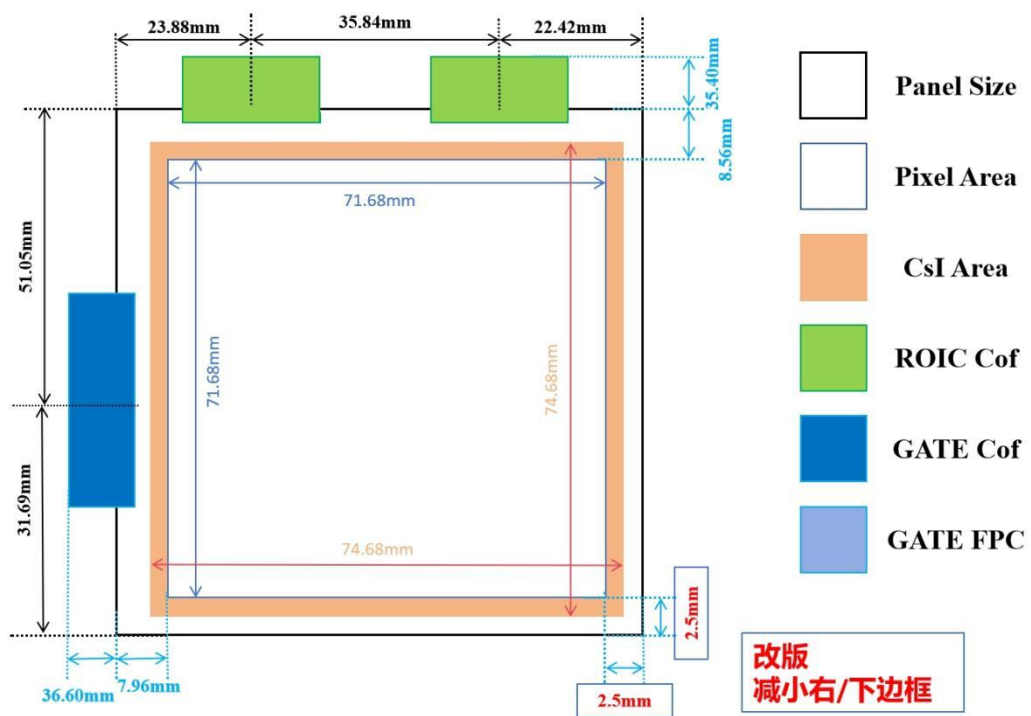
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	Data 2024/11/05	Territoriality Wu Hu	Version 1.0
			Page 6 / 12

5.0 Panel Outline

5.1 TFT Sensor Panel Outline



5.2 Panel After Bonding Outline



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Data 2024/11/05	Territoriality Wu Hu	Version 1.0	Page 7 / 12

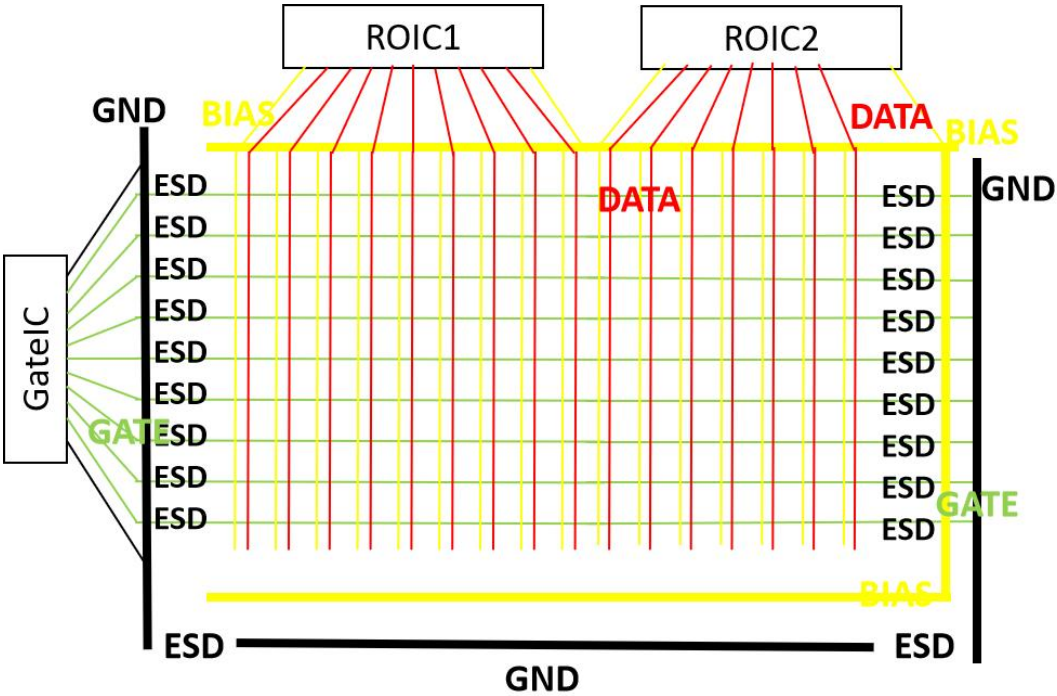
6.0 14 Layers TFT Process of 3.99 inch TFT sensor

There are 14 PEP for 3.99 inch TFT sensor: 4 metal and 1 ITO layers for conducting; 1 IGZO layer and 1 Si layer layers for TFT active layer and photodiode respectively; 5 VIA layers for contact hole; 2 JAS layer for surface morphology flatness.

No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14
layer	M1	ACT	ES	GI	M2	JAS1	PAS1	PXLB	PIN	JAS2	PAS2	M3	ITO	PAS3

7.0 Electrical Diagram of 3.99 inch TFT sensor

Bias line is provided from read out side and gate side, GND line is provide from gate COF side only; ROIC bonding from UP side only, Gate cof bonding from left side only.



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Data 2024/11/05	Territoriality Wu Hu	Version 1.0	Page 8 / 12

7.1 Electrical Input Range of 3.99 inch TFT sensor

	Design parameter	min(V)	typ(V)	max(V)	remark
Electrical	Bias voltage (V)	-10	-5	-3	
	Gate on voltage (V)	/	15	35	
	Gate off voltage (V)	-15	-10	/	

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Data 2024/11/05	Territoriality Wu Hu	Version 1.0	Page 9 / 12

8.0 TEG SPECIFICATION of 3.99 inch TFT sensor

23 items are included in TEG test, and sampling rate are 4 points/sheet for every sheet.

Parameter and Condition

Parameter		Test Condition		SPEC
		Vgs(V)	Vds(V)	
	Vth	6,7,8	10	≤3 V
TFT Characteristic	Mobility	6,7,8	10	≥ 7
	Ron	15	0.5	≤3.5MΩ
	Ioff	-5	0.5	≤ 0.5 fA
Resistance	Rs_M1	L / W = 20000 / 10		≤ 0.15 Ω/□
	Rs_M2	L / W = 20000 / 10		≤ 0.20 Ω/□
	Rs_M3	L / W = 20000 / 10		≤ 1 Ω/□
	Rs_M4	L / W = 20000 / 10		≤ 0.20 Ω/□
	Rs_ITO	L / W = 20000 / 10		≤ 100 Ω/□
Capacitance	GI	100 KHz, +20 V, (1mm²)		1.34E-10 ± 15%
	ES	100 KHz, +20 V, (1mm²)		3.45E-10 ± 15%
	PV1	100 KHz, +20 V, (1mm²)		1.8E-10 ± 15%
	PV2	100 KHz, +20 V, (1mm²)		3.05E-10 ± 15%
	PV3	100 KHz, +20 V, (1mm²)		1.22E-11 ± 15%
	PD	100 KHz, -20 V, (1mm²)		1.10E-10± 15%
Diode Characteristic	PD dark current(1mm²)	Vd = -5 V,		< 7.5 pA

* Vm is the voltage applying on the two pads of TEG

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Data 2024/11/05	Territoriality Wu Hu	Version 1.0	Page 10 / 12

9.0 TEST SPECIFICATION

All TFT array panels would be done by the probe test for electrical properties. All point and line defects in the array are identified by the probe test. The probe test measures the charge of the pixels storage capacitors through TFT under dark environment.

9.1 Defect Definitions

Point defect : A point is a point defect when it’s charge value is 30% higher or less than average value.
Line defect : A defect lines is a row or column that contains more than 5% defective pixels. Pixels on this line will not be included in the defective cluster count.

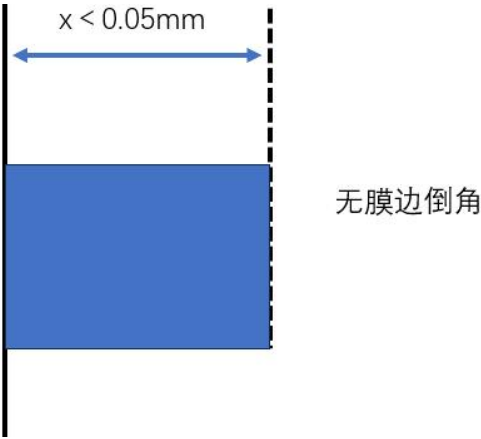
9.2 Defects Specifications

defects specifiactioings	
Global level	
Cluster class_0	320
Cluster class_1	150
Cluster class_2	60
Cluster class_3	30
Cluster class_4	15
Cluster class_5	10
Cluster class_6	5
Cluster class_7	0
Cluster class_8	0
gate line open	5
data line open	5
gap between two gate or data line	10
line shorts	0
T defect	0

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Data 2024/11/05	Territoriality Wu Hu	Version 1.0	Page 11 / 12

10.0 VISUAL INSPECTION & AOI INSPECTION

All TFT arrays are visually inspected. The visual inspection includes an inspection of the array against a white light to identify gross defects such as: The breach angle of glass is not over 90 degree and do not damage the mark on glass edge



- ◆ No scratches and cracks on active area and bonding pad.
- ◆ No surface debris and contamination on active area and bonding pad.
- ◆ No missing material and metal residue on TFT array

